

Project FORESIGHT

AI-Powered Demand & Inventory Intelligence Platform

Project Report

Submitted by	Rinkle
Mentor	Chandan Mishra
Programme	Data Analytics Internship
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GitHub	github.com/rinkle-YADAV

1. Project Overview

ProjectFORESIGHT is a demandforecasting and inventory intelligence platform that helps a retail business plan stock more accurately. It combines data science and machine learning with business intelligence to forecast SKU-level demand, flag stockout and overstock risks, and turn both into ranked, actionable recommendations delivered through an interactive 12-page Power BI dashboard.

Objectives

- Forecast demand and revenue so purchasing can be planned ahead of the season.
- Detect SKUs at risk of stocking out before lead time runs out, so sales are not lost.
- Identify excess and slow-moving stock that ties up cash and adds holding cost.
- Understand what drives revenue: products, categories, seasons, promotions and customer segments.
- Give executives a short, prioritised list of actions with expected impact.

2. Problem Statement

Inventory decisions are usually made on averages and intuition. Too little stock causes lost sales and unhappy customers; too much stock blocks working capital and increases storage cost. Both problems exist together in the same business, in different SKUs and at different times of the year. FORESIGHT answers three questions: what will sell, what will run out, and what is sitting unsold.

3. Data Description

Theanalysiscoversdailytransactions from **1 January 2024 to 31 December 2025**. Key facts from the dataset as used in the dashboard:

Item	Detail
Total revenue	3.10 bn (2024: 1.55 bn, 2025: 1.54 bn)
Gross profit / margin	1.22 bn / about 39.4%
Units sold	511,810
Products and categories	50 products across 5 categories: Home Decor, Furniture, Storage, Kitchen, Lighting
Customers	About 2,000, grouped into High Value, Regular and Occasional segments
Inventory fields	Current stock, reorder point, lead time, inventory value, days until stockout
Calendar fields	Season, quarter, month, weekday / weekend, holiday and promotion flags

4. Technology Stack

Layer	Tools	Purpose
Data preparation	Python (pandas, NumPy), SQL, Excel	Cleaning, joins, feature creation
Modelling	Python (scikit-learn / statsmodels)	Demand forecasting with 95% intervals; risk classification
Analytics and BI	Power BI Desktop, DAX, Power Query	Data model, KPIs, 12-page interactive dashboard
Version control	Git, GitHub	Source code and documentation
Deployment	Streamlit / Power BI Service	Public access to the live project

5. System Architecture

Data flows from raw business tables through cleaning and modelling into a Power BI data model, which feeds the dashboard pages and the final recommendations.



Output: forecast, stockout / overstock flags and ranked executive actions

6. Methodology

6.1 Data preparation. Sales, product master, inventory and customer tables were cleaned (types, duplicates, missing values) and joined on SKU / product name. Calendar features such as season, weekend, holiday and promotion flags were added to explain demand patterns.

6.2 Demand forecasting. Daily revenue was aggregated and forecast with a [MODEL NAME] model. The last part of the history was held out for validation, and accuracy was measured with MAPE ([MAPE VALUE] on the holdout). The dashboard shows the actual line, forecast line and 95% upper and lower bounds, plus a 30-day outlook and forecast by category.

6.3 Stockout risk. Each SKU is compared on current stock versus reorder point, and on days until stockout versus supplier lead time. SKUs are labelled CRITICAL, WARNING or SAFE. CRITICAL means stock will run out before a new order can arrive.

6.4 Overstock and slow-moving stock. Excess stock value, months to clear and annual holding cost are calculated per SKU. A 20% discount scenario estimates the revenue that could be recovered by clearance sales.

6.5 Recommendations. Risk counts, growth and profitability are combined in an executive page with a risk heat map, an impact-versus-difficulty opportunity view and a ranked action list.

7. Dashboard Walkthrough and Insights

ThePowerBIreporthas12pages.Eachpagebelowshowsascreenshot followed by the main findings.

7.1 Executive Dashboard

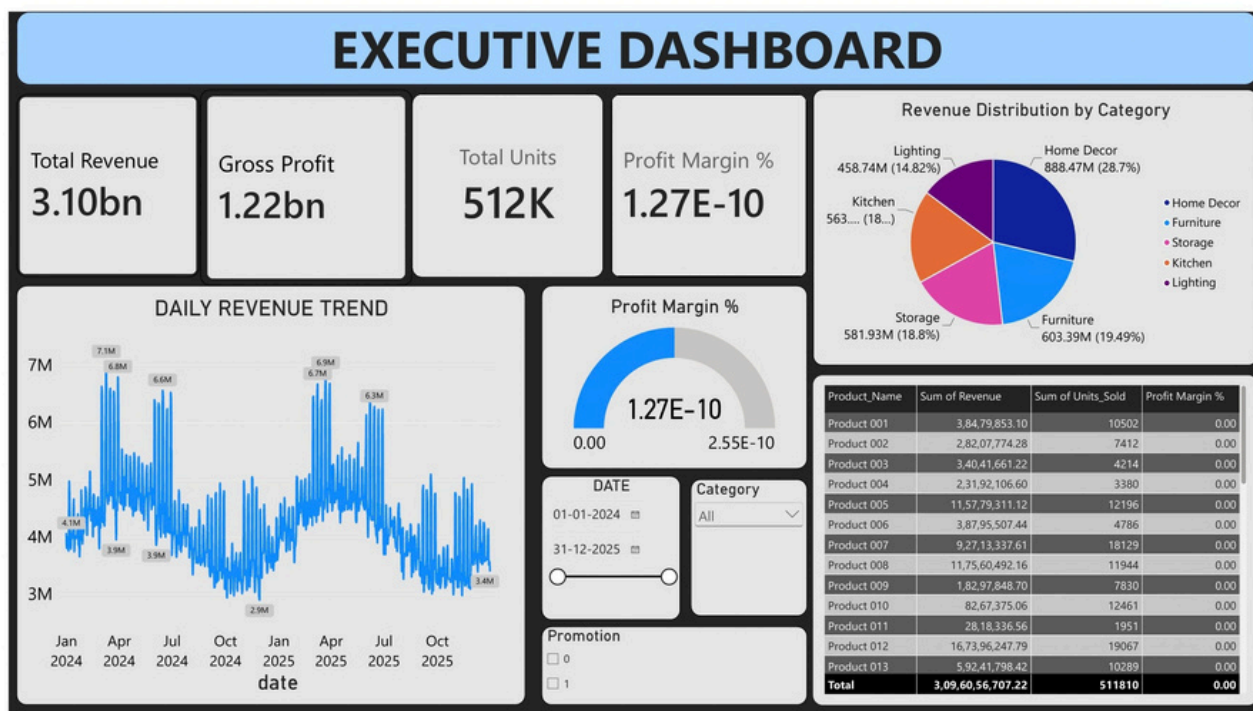


Figure: Executive Dashboard

- Total revenue is 3.10 bn from 511,810 units, with 1.22 bn gross profit.
- Daily revenue peaks at 6 to 7 M in spring (March to June) of both years and falls to about 3 M in October to December.
- Home Decor is the largest category at 28.7% of revenue; Lighting is the smallest at 14.8%.

7.2 Sales Analytics Dashboard

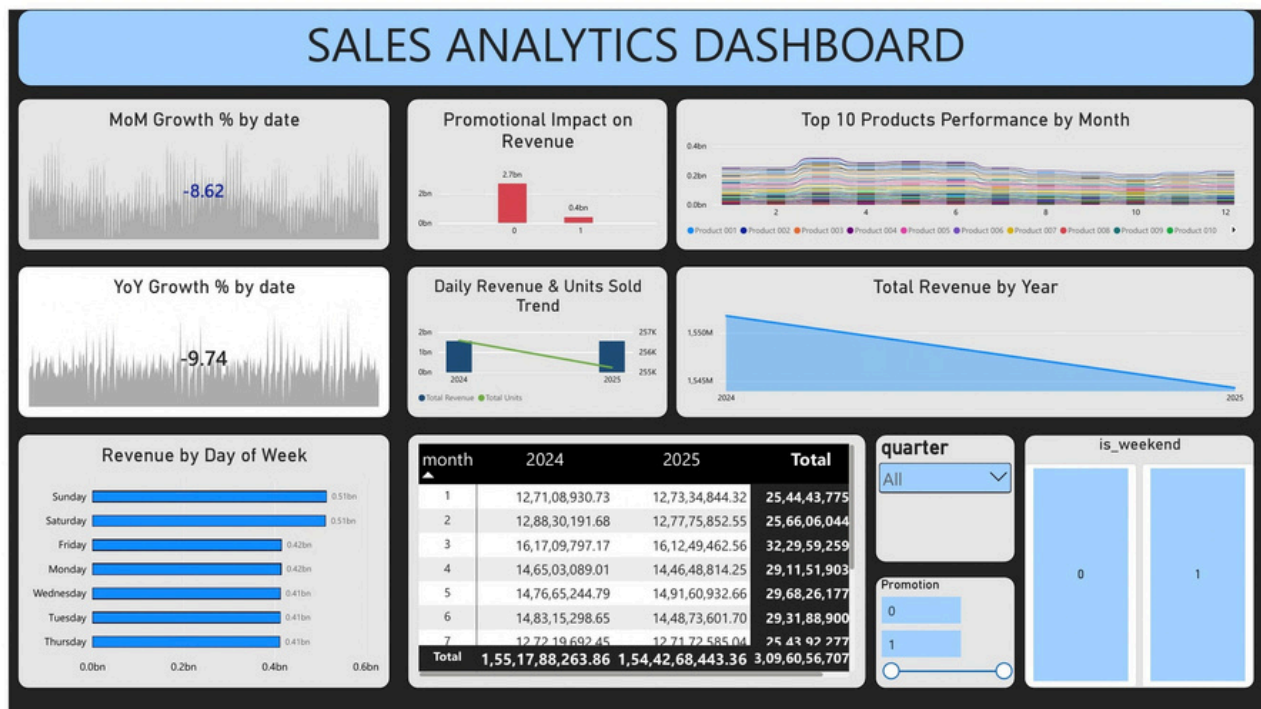


Figure: Sales Analytics Dashboard

- Revenue is almost flat year on year: 1.55 bn in 2024 versus 1.54 bn in 2025 (about -0.5%).
- Saturday and Sunday earn about 0.51 bn each versus 0.41 to 0.42 bn on other days, so weekends bring roughly a third of revenue.
- Non-promotional sales are about 2.7 bn against 0.4 bn for promotional sales. March is the strongest month.

7.3 Product Performance Dashboard

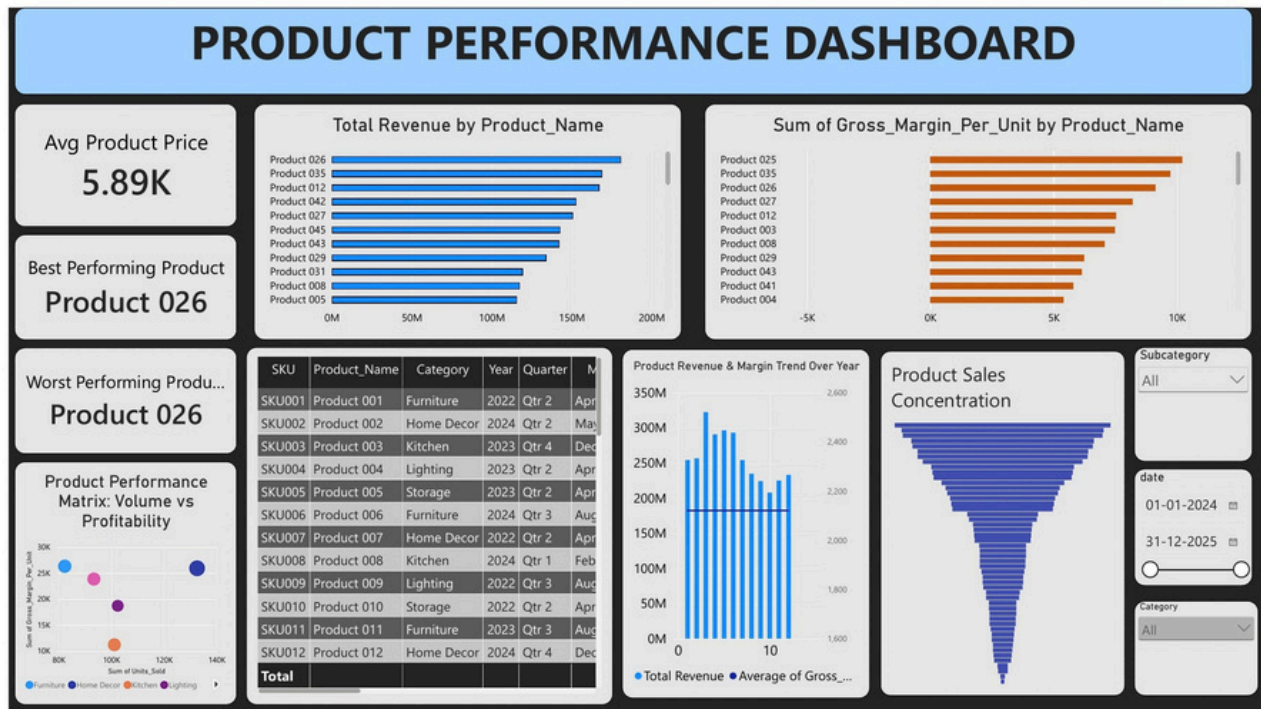


Figure: Product Performance Dashboard

- Product 026, Product 035 and Product 012 lead on total revenue, with Product 026 at around 180 M.
- Gross margin per unit is highest for Products 025, 035 and 026, so some top sellers are also the most profitable.
- The sales concentration chart shows a long tail: a few products drive most revenue while many contribute little.

7.4 Category Performance Dashboard

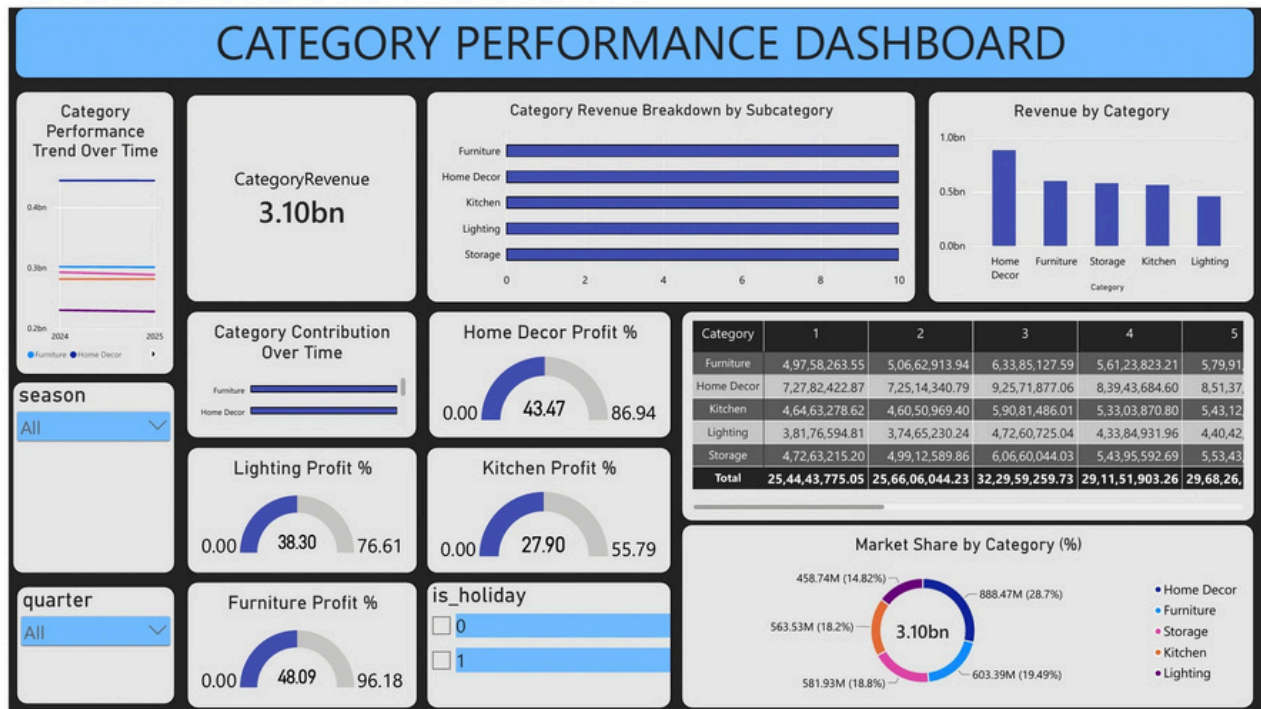


Figure: Category Performance Dashboard

- Revenue split: Home Decor 888 M, Furniture 603 M, Storage 582 M, Kitchen 564 M, Lighting 459 M.
- Furniture shows the highest profit percentage (48.09), followed by Home Decor (43.47) and Lighting (38.30); Kitchen is lowest (27.90).
- Category revenue follows the same monthly shape, peaking in March.

7.5 Inventory Dashboard

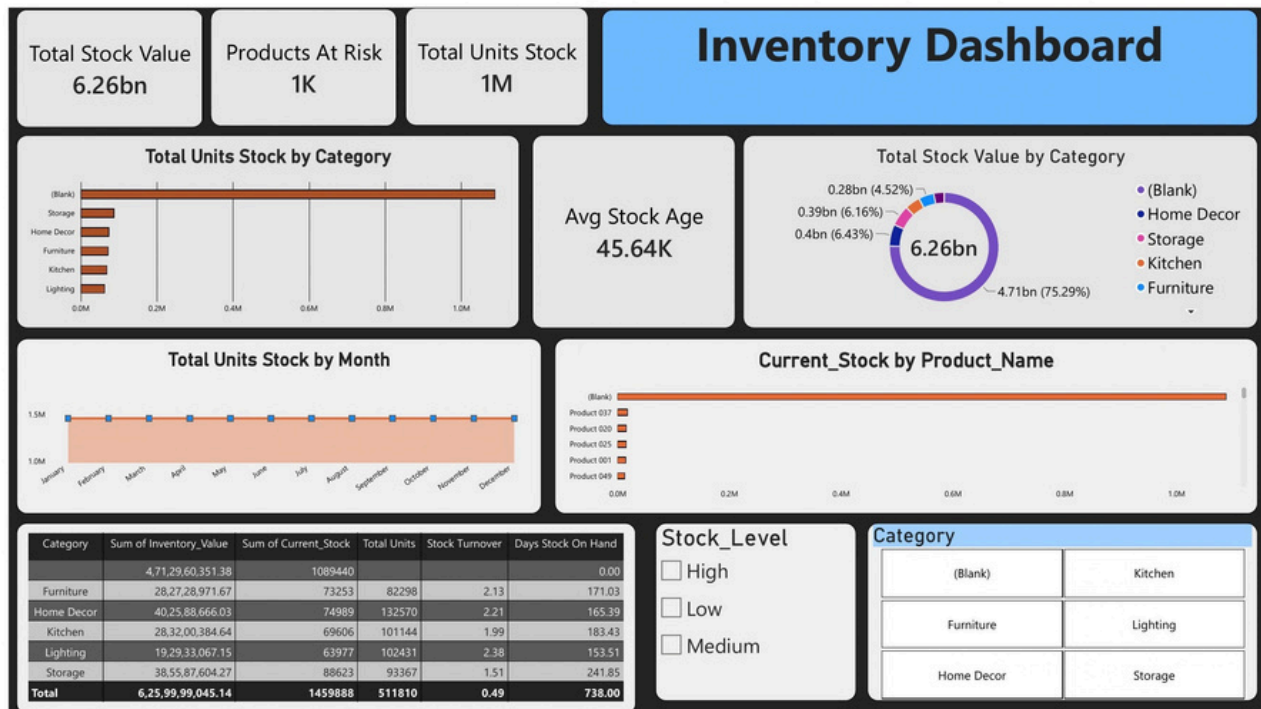


Figure: Inventory Dashboard

- Total stock value is about 6.26 bn, with roughly 1.46 M units on hand.
- Storage holds stock the longest (241.85 days on hand, turnover 1.51); Lighting turns fastest (turnover 2.38, 153.51 days).
- Stock level is nearly constant across months, so inventory is not adjusting to the seasonal demand curve.

7.6 Stockout Risk Dashboard



Figure: Stockout Risk Dashboard

- Risk flags: 727 CRITICAL, 93 WARNING and 3,980 SAFE.
- CRITICAL items cluster at or below zero days until stockout while lead times are 8 to 12 days, so they will run out before replenishment arrives.
- The SKU tables compare current stock with reorder point to show where to reorder first.

7.7 Overstock Dashboard

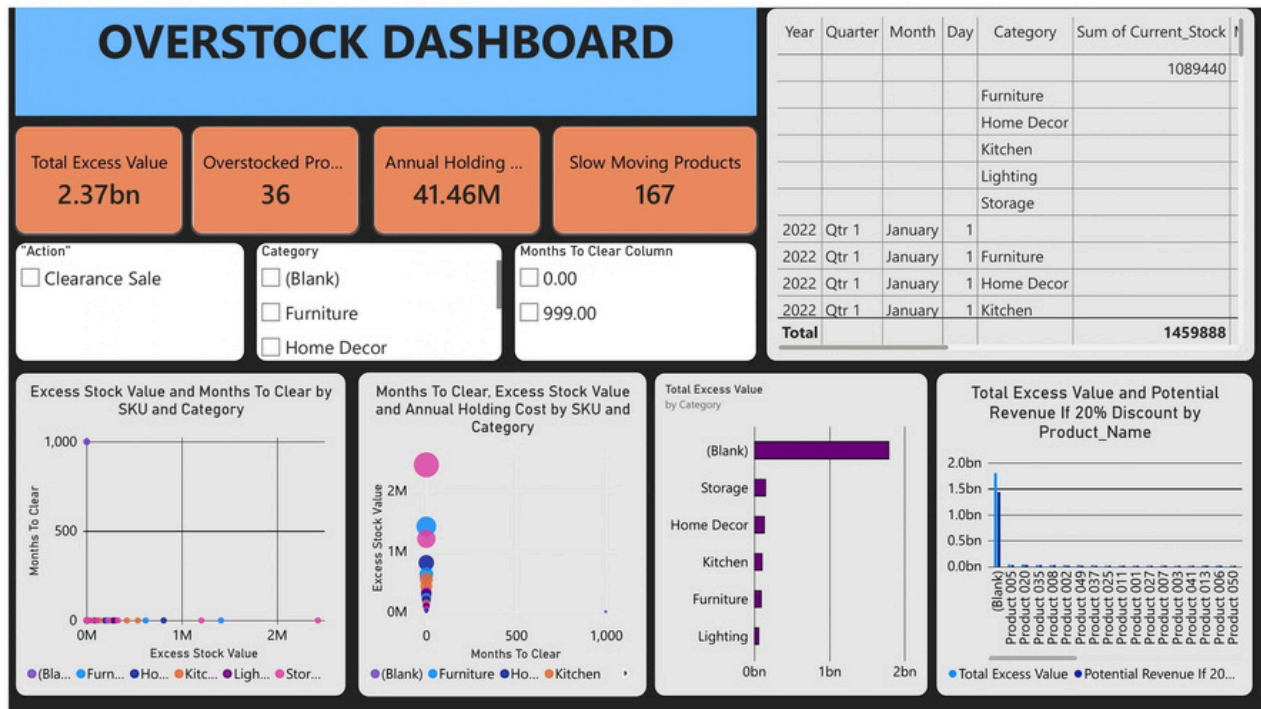


Figure: Overstock Dashboard

- Total excess value is 2.37 bn across 36 overstocked products, with 167 slow-moving products.
- Annual holding cost of the excess is 41.46 M.
- The 20% discount scenario compares excess value with potential revenue to support clearance-sale decisions.

7.8 Promotion Dashboard



Figure: Promotion Dashboard

- Promotional sales are 69,799 units against 442,011 non-promotional units.
- Promo revenue is about 411 M; Home Decor has the highest promotional revenue among categories. In
- the product view, promo revenue rises with promotional units, with some Storage and Home Decor products performing best.

7.9 Seasonality Dashboard

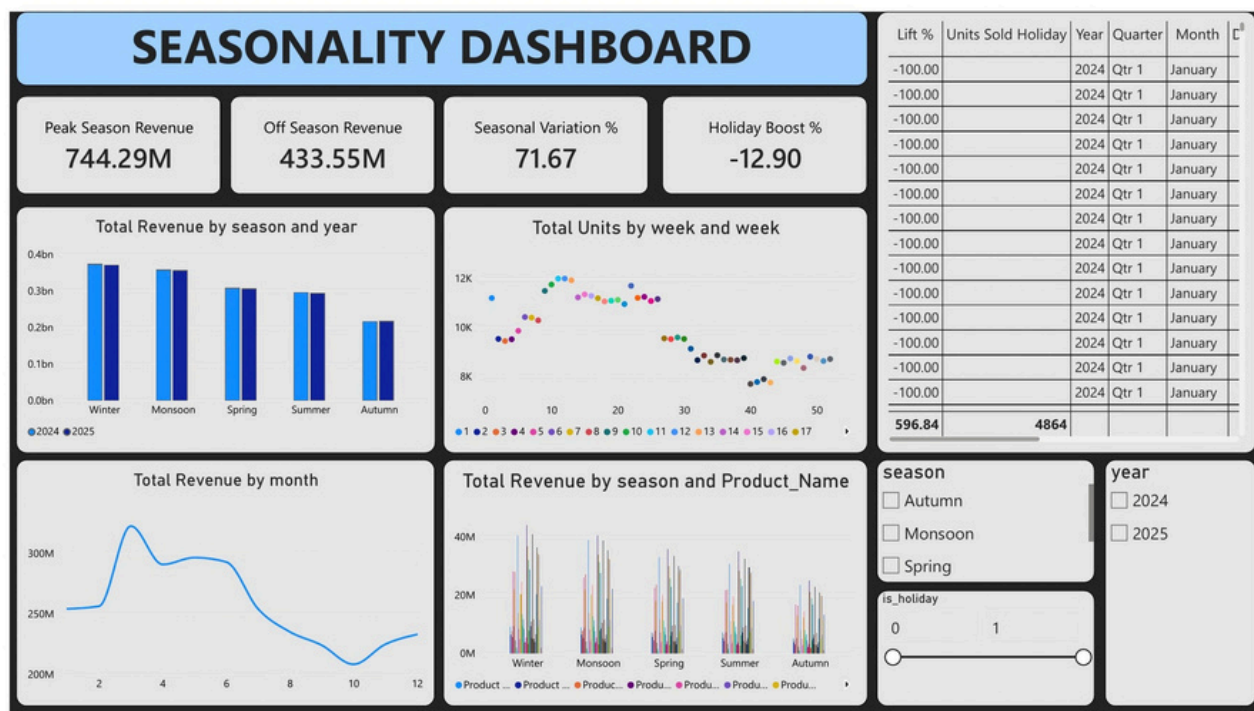


Figure: Seasonality Dashboard

- Peak-season revenue is 744.29 M versus 433.55 M off-season, a 71.67% seasonal variation.
- Winter is the strongest season in both years and Autumn the weakest; October is the lowest month.
- Weekly units are highest in weeks 10 to 14 and lowest late in the year.

7.10 Forecast Dashboard

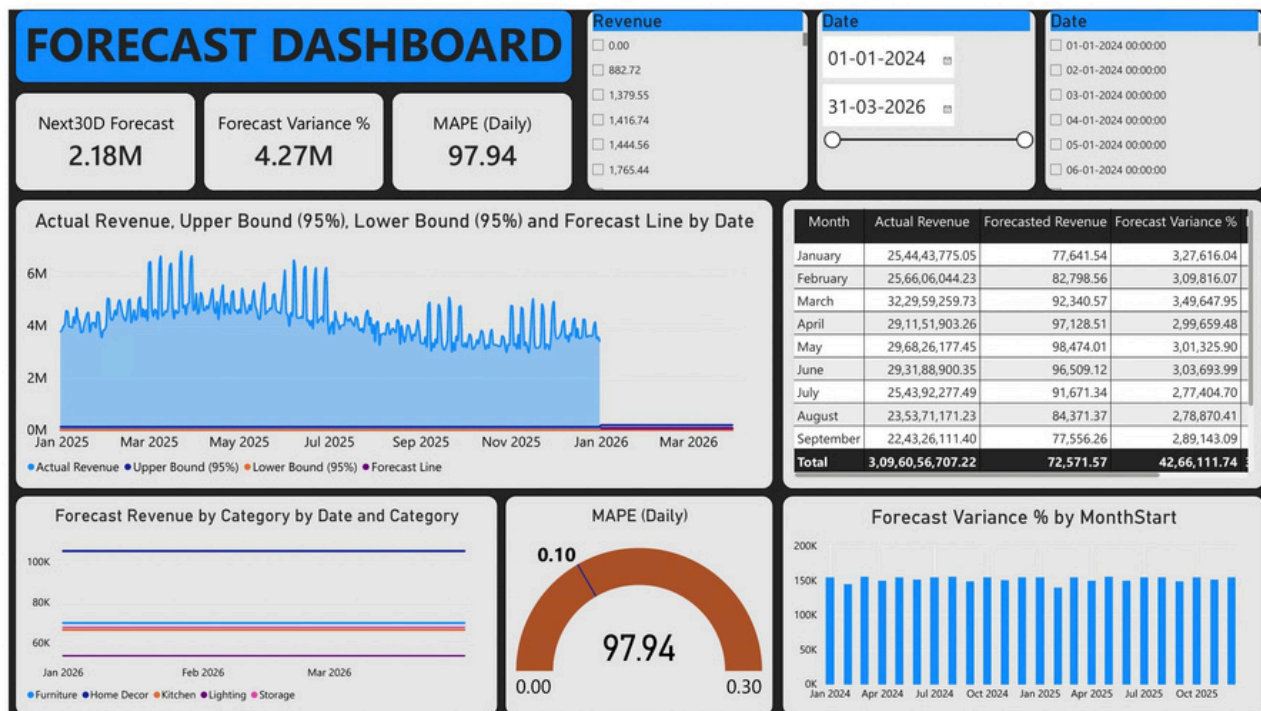


Figure: Forecast Dashboard

- Shows actual revenue with forecast line and 95% upper and lower bounds.
- Includes a 30-day forecast, monthly forecast versus actual comparison, forecast revenue by category and an error (MAPE) gauge.
- Forecasts feed directly into purchasing and reorder planning.

7.11 Customer and Business Insights



Figure: Customer and Business Insights

- About 2,000 customers: 20% High Value (490 customers), 40% Regular and 40% Occasional.
- Regular customers bring the most revenue (about 1.36 bn), High Value about 1.05 bn and Occasional about 0.69 bn. High Value customers give the highest revenue per customer.
- Regular customers buy mostly 11 to 20 times, High Value customers 21 or more times; revenue is spread across all five categories.

7.12 Executive Recommendation Dashboard



Figure: Executive Recommendation Dashboard

- Brings growth, profitability (39.4%), stock turnover (1.4) and stockout risk into one view, with a risk heat map by category.
- The opportunity matrix plots revenue impact against difficulty for customer, inventory, marketing and pricing initiatives.
- Ranked actions: 1) reorder critical products, 2) reduce excess inventory, 3) improve low-performing SKUs, 4) optimise reorder levels, 5) focus on high-growth SKUs.

8. Key Insights and Business Recommendations

Priority	Recommendation	Why	Impact
1 (High)	Reorder critical products now	727 SKUs are flagged CRITICAL	Prevents lost sales
2 (High)	Clear excess stock through targeted discounts	2.37 bn excess value, 41.46 M annual holding cost	Frees cash, lowers holding cost
3 (Medium)	Review low-performing SKUs	Long tail of low-revenue products	Improves sales mix
4 (Medium)	Align reorder levels and stock with seasonality	Stock is flat while demand peaks in spring and winter	Better inventory efficiency
5 (Medium)	Increase stock of high-growth SKUs	Top products lead revenue and margin	Increases revenue

Additional observations: weekends and the March to June period carry the highest demand; Storage is the slowest-moving category; and a small High Value segment contributes a disproportionate share of revenue.

9. Challenges Faced

- Joining inventory records to the product master so every SKU carries the right category.

- Bringing forecast values to the same scale and level (daily total) as actual revenue for a fair comparison.
- Writing correct ratio measures in DAX (margin, growth, retention) that behave properly under filters and totals.
- Designing 12 pages that stay readable while keeping one consistent story for executives.

10. Conclusion and Future Scope

FORESIGHT shows how forecasting and inventory analytics can be combined in one decision tool. It identifies where stock is about to run out, where cash is trapped in excess stock, and which actions should be taken first. Future improvements include SKU-level forecasting models, automatic reorder-quantity suggestions, scheduled data refresh, alerts for critical SKUs, and price and promotion optimisation.

11. Project Links

Deliverable	Link
Source code (GitHub)	[add repository link]
Live deployment	[add public URL]
Demo video	[add video link] [add
Feedback video	video link]